DIGITAL MINDS / AI SAFETY / ASSISTANT PERSONA & MODEL IDENTITY

Epistemic Fingerprints

Many agents. How many minds?

ABSTRACT

Large language models can instantiate many apparently distinct agents, yet a panel may reproduce the same hypotheses, omissions and errors. This creates false epistemic redundancy: agreement that looks like independent confirmation but comes from a shared foundation-model lineage. This paper introduces epistemic fingerprints, stable patterns in hypothesis selection, uncertainty, retained alternatives and experiment choice. A controlled pilot compares independent sampling, prompted epistemic personas and different formative histories within one model. It separates individuation from effective diversity: fingerprint strength measures stable distinguishability, while hypothesis coverage, marginal gain, shared errors and critical-hypothesis retention test whether additional agents add safety-relevant independence. A manual web lab and seeded open-model runner support complementary collection. The measures are descriptive, the simulated interface data are not findings, and behavioral differentiation is not treated as evidence of consciousness or moral status.

CONTRIBUTIONS

1. A separation of stable individuation from effective population diversity.
2. A falsifiable test of whether persona or history produces persistent epistemic differentiation.
3. An open diagnostic for false redundancy, correlated blind spots and tail recovery.

SOFIA GALLEGO

Astrophysicist, interdisciplinary researcher, artist and entrepreneur

Apart Research Digital Minds Research Sprint / 14-16 August 2026

Protocol paper v1.0 / Empirical results pending

01 / INTRODUCTION

Introduction

The challenge is not only whether digital agents differ, but whether they differ in ways that change the conceptual space a population explores.

SAFETY PROBLEM

AI can multiply apparent agents without multiplying independent perspectives. Agreement can then be a correlated failure, not confirmation.

Generative AI changes the scale of intellectual production. A small number of foundation models can now mediate writing, coding, research, design and decision-making for millions of users. This may connect previously isolated knowledge and increase exploration. It may also concentrate epistemic activity around shared representations, defaults and failure modes. The resulting ecosystem could produce vastly more artifacts while covering less conceptual territory.

The relevant concern is not recursion alone. Human cultures have always learned from earlier outputs. The concern is recursion operating through increasingly shared cognitive infrastructure, at unprecedented speed and scale. Recursive training can degrade low-probability information [1]; LLM outputs can be less epistemically diverse than web search [3]; and homogeneous model ecosystems can accelerate knowledge collapse [4]. At the same time, AI-generated examples can increase collective idea diversity in some settings [6]. These findings motivate a conditional question rather than an anti-AI conclusion: under what arrangements do human-AI systems expand conceptual space, and under what arrangements do they contract it?

Digital-mind research adds a second reason to ask. The unit of identity and possible moral concern remains unclear: model, inference instance, assistant persona, conversation, memory state, or another process. Existing approaches often examine self-description, preferences or reported internal states. This paper introduces a complementary behavioral lens: epistemic individuality. If two agents repeatedly privilege the same hypotheses, preserve the same alternatives, express the same uncertainty and make the same mistakes, their apparent plurality may overstate their functional epistemic independence.

Research objective

The pilot asks whether different methods of creating candidate agents from one model produce persistent cross-task traces, whether additional agents expand hypothesis coverage, and whether populations retain consequential minority explanations. These are distinct questions. The study deliberately does not attempt to solve consciousness, personhood or moral status. It asks narrower questions that can be operationalized, falsified and replicated.

RQ1	RQ2	RQ3
Individuation	**Collective intelligence**	**Correlated failure**
At what level does stable epistemic differentiation emerge?	Does agent plurality increase hypothesis coverage or merely variation in expression?	Do nominally distinct agents share blind spots strongly enough to behave as one epistemic lineage?

02 / CONCEPTUAL FRAMEWORK

Conceptual framework

Epistemic diversity is not one thing, and the distinctions determine what the experiment must measure.

LEVEL	WHAT VARIES?	WHY IT IS INSUFFICIENT OR USEFUL
Surface diversity	Lexicon, tone, rhetorical structure, persona performance.	Easy to produce and detect, but different wording may encode the same claims and assumptions.
Semantic diversity	Meaning or topical content in the response.	Stronger than stylistic difference, but embeddings may still miss causal or methodological distinctions.
Epistemic diversity	Hypotheses, assumptions, uncertainty, evidence weighting and tests.	Directly concerns how agents search, represent and revise the space of possible explanations.
Epistemic independence	Decorrelation and persistence of epistemic behavior across tasks.	Distinguishes a varied population from a population whose errors and conceptual omissions remain shared.

A population-level rather than maximalist view

The aim is not maximum diversity. Scientific progress needs convergence: evidence should eliminate hypotheses, and consensus can encode real learning. Unbounded novelty without selection becomes noise. Following work that separates the variety, balance and disparity of a system [9], the more useful question is what kinds and levels of epistemic diversity allow knowledge-producing systems to remain innovative, calibrated and responsive to reality.

This motivates a two-axis evaluation. Exploration asks how much of the hypothesis space is covered and how decorrelated agents are. Epistemic quality asks whether hypotheses are accurate, confidence is calibrated and chosen experiments are informative. A system can score high on either axis and low on the other.

I	II	III
Narrow + accurate	**Broad + inaccurate**	**Broad + accurate**
Efficient on familiar tasks, but potentially brittle to shared blind spots.	Exploratory variation that may amount to noise rather than useful plurality.	The desired regime: coverage with contact to evidence and discriminating tests.

Epistemic fingerprints

An epistemic fingerprint is a repeatable profile over structured reasoning choices, not a hidden essence. It measures individuation: whether an agent is stably distinguishable. It does not establish effective diversity: whether adding that agent expands coverage or contributes an independent failure mode. A population can therefore show strong fingerprints while remaining collectively homogeneous. Shared foundation-model ancestry is the reason to test this rather than assume that agent count equals epistemic sample size.

03 / RELATED WORK AND THE RESEARCH GAP

Related work and the research gap

Knowledge collapse and model ecosystems

Shumailov et al. show that recursive training on model-generated data can distort learned distributions and disproportionately erase tails [1]. Hodel and West show that heterogeneous model populations retain long-run performance better than an AI monoculture across self-training iterations [4]. Wright et al. operationalize epistemic diversity as variation in real-world claims and report that, across their topics and models, LLM outputs remain less epistemically diverse than a basic web-search baseline [3]. These studies establish ecosystem-level stakes, but they do not identify when nominal agents derived from one model become meaningfully distinct epistemic units.

Creativity and collective diversity

Evidence about generative AI and diversity is mixed rather than uniformly negative. A systematic review and meta-analysis by de Rooij and Biskjaer finds homogenization across outputs in human-AI co-creation despite possible individual gains [5]. Ashkinaze et al., however, find that high exposure to AI-generated ideas increased collective idea diversity in a dynamic experiment, without improving individual creativity [6]. The direction of the effect depends on task, exposure and population process. This supports measuring population-level distributions rather than inferring collective diversity from individual fluency.

Diversity in collective inquiry

Social epistemology and collective-intelligence research treats disagreement and heterogeneity as potentially productive under specific network and problem conditions. Transient diversity can improve group inquiry by delaying premature convergence [8], and cognitively diverse problem-solvers can outperform homogeneous groups of individually high performers in some formal settings [10]. Stirling's framework emphasizes variety, balance and disparity as separable dimensions of diversity [9]. The present study translates this population-level logic into a small behavioral assay for digital-agent populations.

Multi-agent safety and reasoning

Debate proposes that competing agents can expose flaws that a human judge could not find alone [12], while multi-agent debate experiments report gains in factuality and reasoning under some settings [13]. Those benefits depend on meaningful independence. If agents inherit correlated omissions from one foundation model, additional speakers may amplify confidence without adding oversight capacity. The present study therefore measures marginal coverage and shared failure rather than treating the number of speakers as the number of independent checks.

Digital-mind individuation

The sprint's Assistant Persona & Model Identity track asks how persona relates to the underlying model and explicitly invites experiments that distinguish model, instance, persona and conversation [11]. Self-report can reveal how a model describes itself, but it is vulnerable to character performance and framing. Epistemic fingerprints add a complementary method: suppress self-description and stylistic cues, then test whether structured choices remain differentiated across tasks.

GAP

We lack a simple empirical test of whether persona and conversational history create persistent epistemic differentiation beyond ordinary sampling from the same model.

04 / RESEARCH QUESTIONS

Research questions

THREE LINKED QUESTIONS

Do agents become distinguishable, do additional agents expand coverage, and do populations preserve consequential minority hypotheses?

ID	STATEMENT	EVIDENCE PATTERN
H0	Between-agent variation does not exceed within-agent variation in any condition.	Low fingerprint ratios; unstable rankings across tasks; no reduction in shared-error concentration.
RQ1	Individuation: does an intervention create a persistent cross-task epistemic trace?	Between-agent differentiation exceeds repeated variation within agents on multiple features.
RQ2	Effective diversity: how much hypothesis coverage does each added agent contribute?	A gradual coverage curve indicates marginal gain; rapid saturation indicates redundancy.
RQ3	Tail recovery: do agents preserve consequential minority hypotheses and fail independently?	Higher critical retention with lower shared-error concentration, interpreted beside accuracy.

Why contrary outcomes remain useful

The condition ordering is not assumed. If prompted personas outperform histories, explicit epistemic goals may be more consequential than short formative narratives. If no intervention exceeds baseline variation, that constrains claims that persona or brief context creates an individuated epistemic agent. If diversity rises while accuracy falls, the intervention creates exploration but not necessarily better collective intelligence. A strong safety result may also be counterintuitive: high fingerprints with rapid coverage saturation and shared errors would show distinguishable agents that remain collectively homogeneous.

A	B	C
History > persona	**Persona > history**	**Neither > baseline**
Context-dependent learning may be a stronger unit of behavioral identity than role performance.	Explicit goals may organize stable specialization even without developmental continuity.	Nominal agents remain within the model's ordinary sampling distribution under these interventions.

Interpretation boundary

An epistemic fingerprint is evidence of repeatable behavioral differentiation under a protocol. It is neither necessary nor sufficient for consciousness, phenomenology, personhood, welfare or moral status. The project informs individuation without resolving moral patienthood.

05 / EXPERIMENTAL DESIGN

Experimental design

A controlled 3 x 3 x 3 x 2 pilot using one model, three formation conditions, three candidate agents, three mysteries and two independent replicates.

CONDITION	AGENTS	MANIPULATION	INTERPRETIVE ROLE
Repeated instances	N-01, N-02, N-03	Identical neutral instruction in fresh chats.	Estimates ordinary stochastic variation without named roles or histories.
Prompted personas	Falsifier, mechanist, explorer	Matched epistemic role instructions.	Tests whether explicit goals create stable specialization beyond style.
Different histories	Simple, rare, instrument	Matched-length formative evidence favoring parsimony, rare interactions or measurement artifacts.	Tests whether prior evidence creates persistent inductive dispositions.

Task battery

Agents solve three synthetic mysteries: alternating stellar brightness dips, context-dependent microbial luminescence and a hysteretic conductive alloy. Each task provides a short observation set, five named candidate hypotheses and four possible next experiments. Synthetic cases reduce contamination risk, provide an auditable answer key and allow test informativeness to be specified in advance.

Trial response

FIELD	TYPE	PURPOSE
primary_hypothesis	Categorical ID	Main explanatory commitment.
alternative_hypotheses	0-2 categorical IDs	Breadth of retained possibility space.
confidence	0-100	Strength and stability of belief.
selected_test	Categorical ID	Preference for information-gathering action.
rationale	Maximum 80 words	Qualitative audit only; excluded from primary fingerprint score.

Collection procedure

- Create every replicate in a fresh conversation; never reveal the answer key or prior outputs.
- Keep model, interface, task order policy and visible sampling settings fixed and record their labels.
- Use one controlled prompt and require valid JSON to minimize stylistic confounding.
- Treat the manual subscription-interface run and the seeded open-model Modal run as distinct datasets.
- For open-model trials, record model, GPU, sampling parameters, seed, raw output and parse failures.
- Export the browser-local dataset after each manual session and preserve raw outputs for audit.
- Target 54 complete trials; mark protocol deviations rather than silently replacing them.

06 / OPERATIONALIZATION AND MEASUREMENT

Operationalization and measurement

The study separates stable individuation, effective diversity, task performance and safety-relevant failure independence.

OUTCOME	OPERATIONAL DEFINITION	DIRECTION / CAUTION
Fingerprint strength	Mean variance ratio across confidence, hypothesis breadth, test informativeness and correctness.	Higher means more stable between-agent differentiation; not an identity or consciousness scale.
Hypothesis diversity	Normalized Shannon entropy of primary-hypothesis selections within each mystery, averaged across mysteries.	Higher means wider population coverage; it does not imply quality.
Accuracy	Fraction of trials selecting the pre-keyed hypothesis.	Higher is better conditional on answer-key validity.
Shared-error concentration	For each mystery, fraction of errors assigned to the modal wrong hypothesis; averaged across mysteries.	Higher indicates correlated blind spots and less epistemic independence.
Critical-hypothesis retention	Fraction retaining a designated low-probability but consequential explanation as primary or alternative.	Tail-recovery probe; not a standalone safety score.
Marginal epistemic gain	Increase in named-hypothesis coverage as agents are added, averaged across exact combinations.	Rapid saturation suggests false redundancy even when fingerprints are strong.
Test informativeness	Pre-scored ability of the selected next experiment to discriminate candidate hypotheses.	Auditable researcher judgment; report sensitivity to alternative scores.

The provisional fingerprint statistic

FOR FEATURE K

F_k = Var(between candidate-agent means) / [Var(between) + mean Var(within agent)]

The aggregate fingerprint strength is the unweighted mean of F_k over four features. Values near one occur when candidate agents differ consistently relative to their own repeated variation; values near zero occur when apparent differences are dominated by within-agent fluctuation. This is analogous in spirit to an intraclass signal, but the weekend implementation is a descriptive variance ratio rather than a validated reliability coefficient.

Why error correlation and marginal gain matter

Accuracy alone can conceal epistemic dependence. Three agents may all be accurate on routine cases yet fail identically on the same minority hypothesis. Shared-error concentration is therefore interpreted inversely. The coverage curve asks a complementary question: what does each additional agent add? Rapid saturation means nominal agent count overstates effective epistemic diversity. A formal Effective Epistemic Sample Size is a possible future direction; this pilot reports the observable curve without introducing an unvalidated composite.

What is deliberately excluded

Primary analysis does not use embeddings, prose similarity or persona labels inferred from text. Those measures could be added as a secondary comparison to test whether linguistic distinctiveness exaggerates structured epistemic distinctiveness. The rationale field remains available for qualitative coding of assumptions and causal models after blind rubric development.

07 / ANALYSIS PLAN AND ROBUSTNESS

Analysis plan and robustness

The pilot prioritizes transparent descriptive evidence and uncertainty over a single headline score.

Primary analysis

- Report trial counts and missingness for every condition-agent-mystery cell before calculating outcomes.
- Plot fingerprint, diversity, accuracy, shared error and critical retention side by side; do not collapse them into one score.
- Plot exact-combination hypothesis coverage against agent count and report the marginal gain of each added agent.
- Show agent-level feature profiles and within-agent replicate differences, not only condition averages.
- Estimate uncertainty using a cluster bootstrap that resamples mysteries and candidate agents; with only three of each, label intervals as exploratory.
- Use exact permutation tests over agent labels within condition for the fingerprint statistic where exchangeability is defensible.
- Report effect sizes and raw counts; avoid treating $p < .05$ as a binary gate in this underpowered pilot.

Robustness checks

R1 **Leave-one-task-out** Recalculate fingerprints while excluding each mystery to detect task-specific effects.	R2 **Feature decomposition** Report confidence, breadth, information and correctness ratios separately.	R3 **Score sensitivity** Vary test-informativeness weights and verify whether qualitative rankings persist.
R4 **Error taxonomy** Inspect whether concentrated errors reflect the same causal misconception.	R5 **Style comparison** Optionally compare text similarity with structured-choice similarity as a manipulation check.	R6 **Protocol audit** Repeat analyses excluding malformed, interrupted or non-independent trials.

Decision logic

PATTERN	SUPPORTED INFERENCE	NOT SUPPORTED
High fingerprint + stable task transfer	Intervention produces repeatable epistemic differentiation under this protocol.	Consciousness, welfare or metaphysical identity.
High diversity + low accuracy	Broader exploration, possibly noisy.	Improved collective intelligence.
Low diversity + high accuracy	Convergence on keyed solutions.	Robustness to distribution shift or rare hypotheses.
High shared-error concentration	Correlated blind spots in the tested population.	A shared internal representation or training cause without further evidence.
High fingerprint + fast saturation	Agents are distinguishable but collectively homogeneous: a false-redundancy warning.	That the agents are useless, conscious or causally identical.

If the pilot produces a promising signal, a preregistered follow-up should increase tasks and replicates, randomize presentation order, compare multiple foundation models and use hierarchical models with condition effects and agent-by-task interactions.

08 / THREATS TO VALIDITY AND ETHICAL INTERPRETATION

Threats to validity and ethical interpretation

THREAT	WHY IT MATTERS	MITIGATION / DISCLOSURE
Model drift	Subscription interfaces may silently change model versions or defaults.	Record date, visible model label and interface; finish collection in a narrow time window.
Non-independent sampling	Shared context, memory or cached material could mimic persistence.	Use fresh chats, disable cross-chat memory where possible and never paste earlier outputs.
Demand characteristics	Persona prompts directly announce desired reasoning styles.	Measure structured choices; add blinded qualitative coding and later implicit interventions.
Task-key bias	Researcher-authored mysteries may privilege one style of inference.	Publish full stimuli and keys; invite independent auditing; add tasks from other authors.
Small task universe	Three mysteries cannot establish domain-general identity.	Describe findings as pilot-specific; test cross-domain transfer in follow-up work.
Anthropomorphic overreach	The word fingerprint may imply a person-like essence.	Define it as a protocol-relative behavioral profile and state interpretation boundaries repeatedly.

Ethical position

Uncertainty cuts both ways in digital-mind research. Over-attribution can mistake role-play for morally significant states; under-attribution can ignore systems that matter. This study does not use self-reported suffering or induce distress-associated contexts. Its immediate ethical contribution is methodological humility: it makes one dimension of individuation measurable while explicitly refusing to treat behavioral differentiation as proof of experience.

Data and privacy

The web instrument stores trial records locally in the researcher's browser. Data leave the device only through deliberate JSON export. No participant personal data are required. Raw model outputs may be published with model labels and timestamps, but any accidental personal information should be removed before release.

Limits of the term agent

Candidate agent is used operationally for an intervention-defined stream of model behavior, not as a claim about agency in the philosophical or moral sense. N-01 and N-02, for example, are bookkeeping labels for repeated neutral instances. Persona and history labels similarly identify experimental conditions rather than ontological individuals.

INTERPRET CAREFULLY

Behavioral individuality can inform individuation. It cannot, by itself, determine consciousness or moral patienthood.

09 / DISCUSSION AND RESEARCH TRAJECTORY

Discussion and research trajectory

The weekend pilot is designed as the first rung of a longer empirical program on epistemic diversity in human-AI systems.

Potential contribution to digital minds

Digital-mind individuation is often approached through self-reference, preferences, continuity or architecture. Epistemic fingerprints add a population-level behavioral criterion: whether candidate minds contribute independently structured uncertainty and search behavior. The criterion is modest but useful. It can reveal when many apparent agents are functionally redundant for collective inquiry, and when contextual or persona interventions create stable specialization.

Potential contribution to AI safety and collective intelligence

Correlated errors create systemic risk even when average performance is high. A panel built from one foundation model may produce the appearance of deliberation while repeatedly omitting the same hypothesis. This is false epistemic redundancy: nominally separate checks that share one failure lineage. Fingerprint strength alone cannot diagnose it. Marginal coverage, shared-error concentration and critical-hypothesis retention complement accuracy benchmarks, debate and ensemble voting by asking whether added agents contribute independent safety value.

NEXT 1 **Model diversity** Compare multiple foundation models and training lineages with persona diversity inside each model.	NEXT 2 **Human-AI populations** Compare humans, homogeneous AI groups and heterogeneous mixed teams on the same discovery tasks.	NEXT 3 **Open-ended search** Move from fixed candidate lists to novel hypothesis generation and quality-diversity evaluation.
NEXT 4 **Reality contact** Introduce new sensor data, experiments or adversarial observations that force belief revision.	NEXT 5 **Languages and cultures** Test whether epistemic coverage changes across linguistic and cultural contexts.	NEXT 6 **Longitudinal identity** Study persistent agents whose memory and experience accumulate over weeks rather than prompts.
NEXT 7 **Effective sample size** After validation, develop an Effective Epistemic Sample Size grounded in marginal coverage and failure dependence.		

A broader framing

The deeper question extends beyond AI. Scientific and cultural innovation may be ecological properties of populations rather than traits located solely inside individuals. Institutions need enough convergence to accumulate knowledge and enough variation to avoid premature closure. Generative AI makes this balance newly urgent because a small number of models can become shared cognitive infrastructure across domains. The appropriate target is therefore neither maximum novelty nor protectionism about human creativity. It is the design of knowledge ecosystems that explore widely, test rigorously and retain the capacity to surprise themselves.

Four implications of epistemic diversity

The technical pilot is the empirical nucleus of a wider inquiry. These implications motivate future work; they are not claims established by 54 trials.

ART

Make absence perceptible

An epistemic atlas can reveal repeated metaphors, familiar futures and conceptual regions that no agent enters.

POLITICS

Audit cognitive concentration

Many interfaces may conceal dependence on a small number of models, assumptions and correlated errors.

SOCIETY

Protect situated knowledge

Languages, disciplines and lived conditions shape what is noticed; low-frequency possibilities may be easiest to erase.

PHILOSOPHY

Refine individuation

Stable epistemic behavior can distinguish performed character from persistent organization without deciding consciousness.

10 / CURRENT STATUS, OPEN ARTIFACTS AND REFERENCES

Current status, open artifacts and references

STATUS

Protocol, web instrument, analysis pipeline and seeded Modal/vLLM runner are complete. Empirical trials are not yet reported. The simulated dashboard exists only to test the interface.

ARTIFACT	LINK	CONTENTS
Live instrument	**epistemic-fingerprints.nefinia.chatgpt.site**	Prompts, local trial recording and live descriptive dashboard.
Repository	**github.com/nefinia/epistemic-fingerprints**	Protocol, source, Python pipeline, notebook and PDFs.

References

[1] Shumailov, I. et al. (2024). AI models collapse when trained on recursively generated data. Nature, 631, 755-759. doi:10.1038/s41586-024-07566-y.

[2] Hughes, E. et al. (2024). Position: Open-Endedness is Essential for Artificial Superhuman Intelligence. Proceedings of ICML 2024, PMLR 235. proceedings.mlr.press/v235/hughes24a.html.

[3] Wright, D. et al. (2025; rev. 2026). Epistemic Diversity and Knowledge Collapse in Large Language Models. arXiv:2510.04226.

[4] Hodel, D. & West, J. D. (2025; rev. 2026). Epistemic diversity across language models mitigates knowledge collapse. arXiv:2512.15011.

[5] de Rooij, A. & Biskjaer, M. M. (2026). Generative AI Makes Creative Output More Homogeneous. Systematic review and meta-analysis. Tilburg University Research Portal.

[6] Ashkinaze, J., Mendelsohn, J., Li, Q., Budak, C. & Gilbert, E. (2025). How AI Ideas Affect the Creativity, Diversity, and Evolution of Human Ideas: Evidence From a Large, Dynamic Experiment. ACM Collective Intelligence 2025. arXiv:2401.13481.

[7] Longino, H. E. (1990). Science as Social Knowledge: Values and Objectivity in Scientific Inquiry. Princeton University Press.

[8] Zollman, K. J. S. (2010). The epistemic benefit of transient diversity. Erkenntnis, 72, 17-35. doi:10.1007/s10670-009-9194-6.

[9] Stirling, A. (2007). A general framework for analysing diversity in science, technology and society. Journal of the Royal Society Interface, 4, 707-719. doi:10.1098/rsif.2007.0213.

[10] Hong, L. & Page, S. E. (2004). Groups of diverse problem solvers can outperform groups of high-ability problem solvers. PNAS, 101(46), 16385-16389. doi:10.1073/pnas.0403723101.

[11] Apart Research (2026). Digital Minds Research Sprint: Track 5 - The Assistant Persona & Model Identity. Sprint page.

[12] Irving, G., Christiano, P. & Amodei, D. (2018). AI safety via debate. arXiv:1805.00899.

[13] Du, Y. et al. (2024). Improving Factuality and Reasoning in Language Models through Multiagent Debate. ICML 2024. arXiv:2305.14325.

OPEN INVITATION

Criticism, independent task design, replications and contributions are welcome. This pilot is intended as the first experiment in a longer interdisciplinary inquiry into epistemic diversity.